

Web Crawler

Functional requirements:

user should be able to download file metadata,

Our Distributed system should have a retry mechanism for faulty messages,

Non-functional requirements;

Data-partitioning,

load balancing,

fault tolerance,

vertical scalability,

horizontal scalability,

low latency,

high throughput,

CAP theorem (we need to prioritise 99.99% uptime, i.e Availability >> Consistency)

Core Entities:

user,

url,

file metadata,

API :

POST:/click

{

 userID,

 url

}

HLD :

